Name: _____

Date: _____

Mr. Rawson • Y9 MYP Physics

2.1 – Circuit Diagrams & Symbols

Unit 2: Electricity & Energy

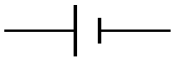
The question: *how do you tell someone else exactly how to build your circuit, without drawing a picture of every wire and battery?* Electricians, engineers and physicists all use the same trick: a small set of **symbols**, agreed everywhere, standing in for the real components. Learn the symbols and you can read — and build — any circuit in this unit.

Learning intentions

All students will learn the symbol, name and job of: wire, cell, switch, bulb, resistor — and build and draw a simple circuit from a diagram.
Most students will also use a **variable resistor** to control a circuit.
Some students will also meet the **LED** and the **capacitor**.

Part 1 — The symbols everyone needs

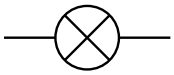
Five components, five symbols. Learn the shape, not just the name — in a diagram you will only ever see the shape.



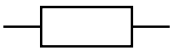
Cell
pushes charge round the circuit



Switch
opens and closes the loop



Bulb
lights up when current flows



Resistor
resists the flow of current

A plain straight line is the **wire** — there is no separate symbol for it, because the wire *is* the line joining everything else together.

1) **Label** the four symbols below with their names, from memory — no peeking at the row above.



Model answer: *Left to right: resistor, switch, cell, bulb.*

Build it

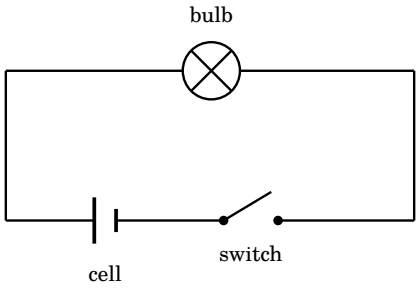
Take a cell, a switch, a bulb and three wires. Build the circuit so the bulb lights when the switch is closed. **Predict first, then test:** what happens to the bulb when you open the switch?

Check: the bulb lights with the switch closed, and goes out the instant you open it. If it does not, you have a gap in the loop — follow the wire all the way round and find it.

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Part 2 — Reading and drawing a circuit diagram

A **circuit diagram**, also called a **schematic diagram**, is the real circuit, redrawn as a neat loop of symbols. Here is the circuit you just built, drawn properly:



Schematic diagram rules

1. *A circuit must be a complete loop. Current flows only when there is an unbroken path from the cell, all the way round, and back to the cell.*

2. *The length of the wires and the shape of the loop make no difference to the circuit. That is why any circuit can be redrawn as a neat rectangle.*

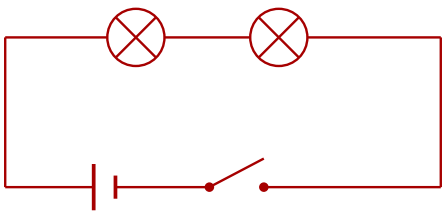
3. *Every circuit needs a source that **provides** electrical energy (a cell or a battery) and at least one component that **uses** it (a bulb, a resistor, a motor). A cell joined by nothing but wire is a **short circuit**. The wire gets hot and the cell quickly runs flat.*

4. *Current from a cell flows in **one direction**, out of one end of the cell, all the way round the loop, and back into the other end.*

Good circuit diagrams

- Use a ruler and a sharp pencil.
- Draw the loop as a neat rectangle.
- Keep every symbol roughly the same size.
- Follow the loop all the way round before you lift your pencil.

2) **Draw** the circuit diagram for: a cell, a closed switch, and **two** bulbs, one after another in a single loop. Use the rules above.



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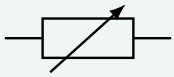
Part 3 — Adding a resistor

A **resistor** does one job: it resists the flow of current, on purpose, so less current flows than without it.

3) **Predict** what happens to the brightness of a bulb if you add a resistor into the same loop, in series with it. Then explain *why*, using the word **resist**.

__Model answer:__ The bulb gets dimmer. The resistor resists the current, so less current flows round the whole loop — and a dimmer bulb is what less current through it looks like.

Most students — the variable resistor



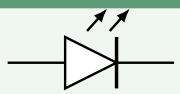
Variable resistor
a resistor you can turn up or down

4) **Draw** the circuit diagram for: a cell, a closed switch, a bulb, and a variable resistor, all in one loop. Then state what turning the variable resistor's knob *up* does to the bulb.

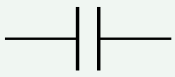
Diagram: the same rectangle loop as Part 2, with the variable-resistor symbol in series anywhere in the loop, in place of one bulb.

__Model answer:__ Turning it up increases the resistance, so less current flows and the bulb gets dimmer.

Some students — LED and capacitor



LED
lights up, but current can only flow one way through it



Capacitor
stores charge, then releases it

5) **Explain** the one thing an LED can do in a circuit that an ordinary bulb cannot.

__Model answer:__ An LED only lets current flow through it in one direction — put it in the circuit the wrong way round and it will not light, however the switch is set. An ordinary bulb lights either way round.

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Part 4 — On your own

A torch contains a cell, a switch, and a bulb, connected in a single loop.

6) **Draw** the torch's circuit diagram, labelling each symbol. Then add a **variable resistor** into the same loop, so the torch's brightness can be turned up and down.

The Part 2 loop (cell, switch, bulb), with a variable-resistor symbol added anywhere else in the loop, in series.

HOMEWORK

1. Draw and label the symbol for each of the five components from Part 1, from memory.
2. Find **one** real circuit at home or at school (a torch, a lamp, a remote control …) that uses at least three of today's components. Name the object and list the components it uses.

Hand in: your five symbols and your one real example, next lesson.